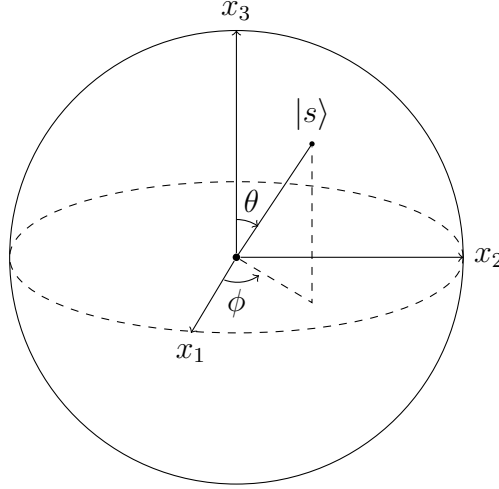


Bloch sphere

Let $|s\rangle$ be the spin state

$$|s\rangle = \begin{pmatrix} \cos(\theta/2) \\ \sin(\theta/2) \exp(i\phi) \end{pmatrix}$$

Spin $|s\rangle$ can be visualized as a point on a unit sphere.



Start with an arbitrary complex vector $|\chi\rangle$ such that $\langle\chi|\chi\rangle = 1$.

$$|\chi\rangle = \begin{pmatrix} A \exp(i\phi_1) \\ \sqrt{1 - A^2} \exp(i\phi_2) \end{pmatrix}$$

Let $|s\rangle$ be the spin

$$|s\rangle = \exp(-i\phi_1) |\chi\rangle = \begin{pmatrix} A \\ \sqrt{1 - A^2} \exp[i(\phi_2 - \phi_1)] \end{pmatrix}$$

Substitute $\cos(\theta/2)$ for A and $\sin(\theta/2)$ for $\sqrt{1 - A^2}$.

$$|s\rangle = \begin{pmatrix} \cos(\theta/2) \\ \sin(\theta/2) \exp[i(\phi_2 - \phi_1)] \end{pmatrix}$$

By equality of expectation values $|\chi\rangle$ and $|s\rangle$ are representations of the same spin.

$$\begin{aligned} \langle S_x \rangle &= \langle \chi | S_x | \chi \rangle = \langle s | S_x | s \rangle \\ \langle S_y \rangle &= \langle \chi | S_y | \chi \rangle = \langle s | S_y | s \rangle \\ \langle S_z \rangle &= \langle \chi | S_z | \chi \rangle = \langle s | S_z | s \rangle \end{aligned}$$

Hence every $|\chi\rangle$ has a place on the Bloch sphere.

Eigenmath script